

# Build-your-own computers

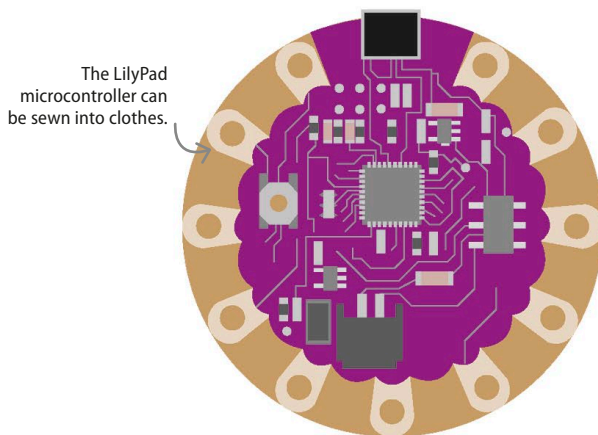
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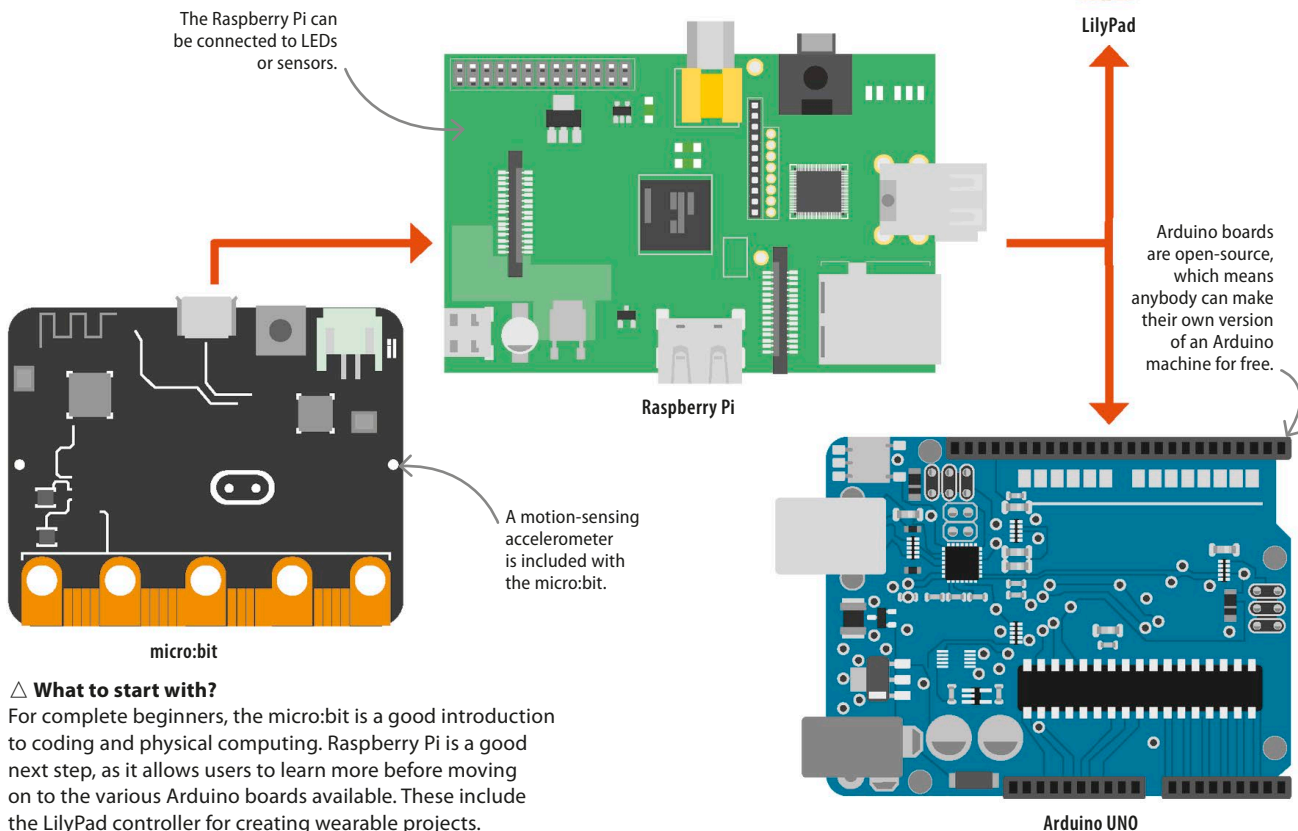
Most computers today are slick consumer devices that hide what's going on inside them. Several organizations are challenging this by encouraging people to build their own machines.

## Equipment required

Do-it-yourself (DIY) computers are split into low-cost microcomputers, such as the Raspberry Pi, and microcontrollers, devices that hold only one program at a time. Although the devices themselves are inexpensive, they all require additional equipment. A Raspberry Pi needs a keyboard, mouse, HDMI cable, screen, power supply, and SD card. All the required software is free via the Raspberry Pi site. Microcontrollers, such as the micro:bit and Arduino boards, need another computer where the code to be uploaded to them can be written and transferred to the microcontroller via a USB cable.



The LilyPad microcontroller can be sewn into clothes.



### △ What to start with?

For complete beginners, the micro:bit is a good introduction to coding and physical computing. Raspberry Pi is a good next step, as it allows users to learn more before moving on to the various Arduino boards available. These include the LilyPad controller for creating wearable projects.